

JAZZ VOICE CORES

VOIP PHONE PROCESSORS



Telephone Handsets



Desktop Phone Terminals



Multi-Function Devices



Embedded VoIP

Highly efficient cores enabling rapid integration of complete speech processing.

The Jazz VOIP Phone Processor (VPP) cores are highly optimized solutions specifically tailored to the evolving requirements of voice-over-IP handsets. There are increasing opportunities for integrating VoIP handsets into traditional equipment as well as new integrated solutions (digital TVs, portable media players, etc.) The Jazz VPP cores can get you to market quickly and efficiently. They support a wide range of speech standards and are optimized for the high quality speech processing at a low cost. The VPP cores are part of the broad Jazz Voice family from Improv.

Voice-Over-IP functionality has found its way into the home, into businesses and into portable devices. The popularity of IP telephony through service providers, Vonage and Skype has increased the market significantly for new products. The Jazz VPP solutions support a complete set of speech and telephony software modules that can be loaded from external memory; this allows a single core to support a broad range of standards.

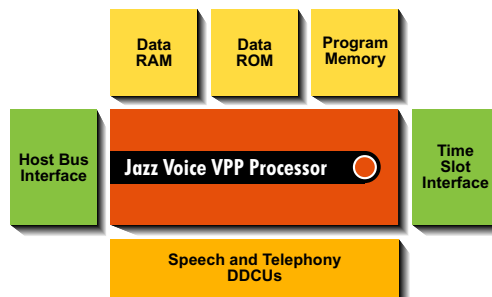
The Jazz VPP cores provide a complete package of telephony and speech processing. The cores are pre-configured and pre-optimized and come with a complete suite of telephony software modules. The cores have built-in interfaces for standard on-chip buses and memory management to allow direct drop-in into an existing SoC architecture.

OPTIMIZED CORES

The Jazz VPP cores can be used as a black box implementation of a complete voice processing subsystem on a chip. Each core can operate as a standalone block or as a slave to a host processor.

Each VPP core has a single, application-specific processor that combines a general-purpose instruction set with application-specific instructions to accelerate telephony processing. Each core contains a very-long instruction word (VLIW) processor built using Improv's Jazz Composer tools.

In addition to the processor, the hardware core contains embedded memory, a host bus interface (HBI) that can connect to most standard on-chip buses including AMBA, OCP and Wishbone, and an optional Time Slot Interchange (TSI) block that allows direct integration with SLIC/SLAC interfaces.



KEY FEATURES

- Complete silicon-ready solution
- Optimized for multiple standards and multiple channels
- Complete telephony framework for a drop-in silicon solution
- Available in multiple configurations
- Fully synthesizable and foundry independent
- Supported by a comprehensive software development tool suite
- Optionally available as configurable cores for additional customization

VPP SOFTWARE MODULES

The VPP telephony software modules have been designed to fully exploit the unique advantages of the VPP architecture.

- G.711 Speech Codec
- G.726 Speech Codec
- G.723.1 Speech Codec
- G.728 Speech Codec
- G.729 a/b Speech Codec
- G.165/G.168 Line Echo Cancellation
- Comfort Noise Generation and Voice Activity Detection
- Caller-Id
- Fax Detection and Pass-Through
- Dial-Tone Management and Call Progress Tones
- Conferencing
- V.18 Hearing Impaired Support

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AVAILABLE PRODUCTS

The Jazz Voice VPP family consists of the following specific products:

- **VPP Embedded 2 Channel Core** is optimized for two channels of full duplex speech codecs. It is targeted for embedded integration into devices not requiring an analog phone jack.
- **VPP Home 2 Channel Core** is optimized for two channels of full duplex speech codecs as well as integration with SLIC/SLAC interfaces and line echo cancellation.
- **VPP Desktop 4 Channel Core** is optimized for four channels of full duplex speech codecs, full telephony and integrated support for conferencing.

IMPLEMENTATION CHARACTERISTICS

	<u>GATE COUNT</u>	<u>CLOCK</u>	<u>POWER</u>
VPP Embedded 2-Ch	184K gates	20 MHZ	15 mW
VPP Home 2-Ch	208K gates	40 MHZ	25 mW
VPP Desktop 4-Ch	184K gates	90 MHZ	60 mW

These numbers represent the maximum requirements for running the VPP Telephony Software modules in a TSMC 0.13um process. Low power is a critical component of a successful chip for these markets. The Jazz Voice VPP cores have been highly optimized for high performance at low clock rates and low power.

- Fully synchronous design
- Silicon-proven in 0.18um and 0.13um technologies
- Advanced power management features
- Hardware Debug Environment through JTAG port
- Support for prioritized interrupts
- Full scan with BIST for test
- Boot-loadable firmware and smart caching (Jazz Jukebox)
- No custom physical components required

DEVELOPMENT TOOLS

A complete software development environment called the Jazz Standard tool suite supports Jazz Voice cores. This tool suite includes compiler, debugger, assembler, cycle-accurate instruction-set simulator and profiler/analyzer. The environment runs on Windows XP Professional, Windows 2000 and Windows NT environments.

The cores also come with a complete implementation environment called the Jazz Generator which creates complete synthesizable RTL, EDA tool scripts, verification environment and FPGA mapping.

As a separately purchasable item, Improv offers a development platform using Improv's FPGA-Based Jazz Rehearsal cards with a full verification environment to enable verification and system integration. These boards are augmented by the Jazz Rehearsal Telephony Daughtercard allowing direct, real-time testing of telephony features.

DELIVERABLES

- VPP Processor Core
- VPP Telephony Software Modules
- Verification Environment
- Jazz Standard Tool Suite
- Jazz PSA Generator
- User's Guide
- Implementation Guide
- Programmer's Guide
- Jazz Composer Tool Suite (for configurable package only)

LICENSING INFORMATION

The Jazz Voice VPP cores are available today and can be licensed from Improv Systems. For additional information regarding the Jazz Voice family and other licensable Improv technology, please contact Improv Systems at sales@improvsys.com, or visit our website at <http://www.improvsys.com>. A list of worldwide local representatives can be found on the website.

ABOUT IMPROV SYSTEMS, INC.

Improv Systems, Inc. develops and licenses pre-configured and configurable Jazz Media, Jazz Voice and Jazz DSP cores. Improv's Jazz processor is a configurable DSP that allows designers to create optimized DSP cores targeted to consumer electronics and telecommunications markets. For more information, please visit www.improvsys.com.

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